



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF SCIENCE

Postgraduate Programme

(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED

“Master of Science”

(Botany)

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Dean faculty of Science, Bhabha University

**Master of Science
(M.Sc.-Botany)
I-IV Semester**

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester Pattern)

First Semester

Course No.	Name of the Course	Total
MBT 101	Biology & Diversity of Viruses, Bacteria and Fungi	100
MBT 102	Biology & Diversity of Algae, Bryophytes and Pteridophytes	100
MBT 103	Biology & Diversity of Gymnosperms	100
MBT 104	Plant Ecology	100
MBT 105	Practical I – based on Course PG 101 & 104	50
MBT 106	Practical II – based on Course PG 102 & 103	50
	Total	500

Second Semester

Course No.	Name of the Course	Total
MBT 201	Plant Development & Reproduction	100
MBT 202	Morphology & Taxonomy of Angiosperms	100
MBT 203	Utilization & Conservation of Plant Resources	100
MBT 204	Cell Biology of Plants	100
MBT 205*	Practical I – based on Course PG 201 & 202	50
MBT 206	Practical II – based on Course PG 203 & 204	50
	Total	500

*N.B.: PG 205 will include the following points:

1. Numbers of representative families (about 25) shall be taken up in the practical classes describing the plants up to species level.
2. Study of the primary and secondary anatomy of root and stem in dicots and monocots.

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M.Sc. Botany (Semester Pattern)

Third Semester

Course No.	Name of the Course	Total
MBT 301	Plant Physiology	100
MBT 302	Plant Biochemistry & Metabolism	100
MBT 303	Cytogenetics & Genetics	100
MBT 304	Molecular Biology of Plants	100
MBT 305	Practical I – based on Course PG 301 & 302	50
MBT 306	Practical II – based on Course PG 303 & 104	50
	Total	500

Fourth Semester

Course No.	Name of the Course	Total
MBT 401	Plant Cell, Tissue & Organ Culture	100
MBT 402	Biotechnology & Genetic Engineering	100
MBT 403	Limnology	100
MBT 404	Elective*	100
MBT 405	Practical I – based on Course PG 401 & 402	50
MBT 406	Practical II – based on Course PG 403 & 404	50
	Total	500

- * MBT 404[I] Industrial Microbiology
MBT 404[II] Taxonomy of Angiosperms
MBT 404[III] Bioinformatics and Computer Application & Biostatistics

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

First Semester

Course MBT 101: Biology & Diversity of Viruses, Bacteria and Fungi

- UNIT I: Viruses: characteristics and ultrastructure of virions, isolation and purification of viruses; chemical nature, replication, transmission of viruses; economic importance.
- UNIT II: Archaeobacteria and Eubacteria: General account; ultrastructure, nutrition and reproduction; biology and economic importance; cyanobacteria – salient features and biological importance.
- UNIT III: Classification of bacteria, Actinomycetes, *Mycoplasma*, *Rickettsiae*, *Chlamydiae* and their significance.
- UNIT IV: Mycology: classification and general characters of fungi; substrate relationship in fungi; cell ultrastructure; unicellular and multicellular organization; cell wall composition; nutrition (saprobic, biotrophic, symbiotic); reproduction (vegetative, asexual, sexual), heterothallic; parasexuality; recent trends in classification.
- UNIT V: Phylogeny of Fungi: Phylogeny of fungi; general account of Mastigomycotina, Zygomycotina, Ascomycotina, Basidiomycotina, Deuteromycotina; fungi in industry, medicine and as food; fungal diseases in plants and humans; Mycorrhiza; fungi as biocontrol agents.

Suggested Readings

- Alexopoulos, C.J. Mims, C. W. and Blackwell, M; 1996: Introductory coO' Mycology, Jhon Wiley & Sons Inc.
- Clifton, A; 1958: Introduction to Bacteria, McGraw- Hills Book Co. New Delhi.
- Madigan, M T. Martinko, J. M and Parker Jack; 1997: Brock Biology Of Microorganisms, (8th edition) Prentice Hall, N.J. U.S.A
- Mandahar, C. L.; 1978: Introduction to Plant Viruses. Chand & Co. Ltd. Delhi.
- Mehrotra, R.S. and Aneja, R.S.; 1998: An Introduction to Mycology. New Age Intermediate Press.
- Rangaswamy, G. and Mahadevan, A; 1999: Diseases of Crop Plants in India (4th edition). Prentice Hall of India Ltd. New Delhi.
- Webster, J.; 1985: Introduction to Fungi Cambridge University Press. Dubey, R C. & Maheshwari, D. K.; 2005: A Text Book of Microbiology, S. Chand Publisher, New Delhi

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

First Semester

Course MBT 102: Biology & Diversity of Algae, Bryophytes and Pteridophytes

- UNIT I: Algae in diversified habitats; thallus organization; cell ultrastructure; reproduction; criteria for classification of algae, pigments, reserve foods, flagella; classification.
- UNIT II: Salient features of Protochlorophyta, charophyta, chlorophyta, xanthophyta, bacillariophyta, phaeophyta and rhodophyta; algal blooms; algal biofertilizers; algae as food, feed and industrial uses.
- UNIT III: Morphology, structure, reproduction and life history of bryophyta; distribution, classification, general accounts of marchantiales, jungermanniales, anthocerotales, sphagnales, funariales and polytrichales; ecological and economic importance.
- UNIT IV: Morphology, anatomy, reproduction and life history of pteridophyta; classification, evolution of stele, heterospory and origin of seed habits.
- UNIT V: Introduction to psilopsida, sphenopsida and pteropsida.

Suggested Readings

- Smith G. M.~ Cryptogamic Botany Vol I (2nd edition)~ TataMcGraw-Hill Publishing Company Ltd. Bombay -New Delhi.
- Kumar H. D. 1988: Introductory Phycology. Affiliated East-West Press Ltd. New Delhi.
- Paribar~ N.S. 1991: Bryophyta. Central Book Depot. Allahabad.
- Brower~ 1926: Primitive Land Plants~ Cambridge At the University Press.
- Kashyap~ 1972 Live Worts of Western Himalayas and Punjab. Researchco Publication.
- Smith, G. M.~ Cryptogamic Botany Vol n (2nd edition)~ TataMc Graw -Hill Publishing Company~ Bombay -New Delhi.
- Puri P. 1980~ Bryophyta -Morphology, Growth & Differentiation. Atma Ram & Sons, Delhi.
- Chopra & Kumar~ 1988: Biology of Bryophyta; Wiley Eastern Ltd.
- Ram Udar; 1970: An Introduction to Bryophyta; Shashidhar Malviya Prakashan
- Watson; 1968: Structure and life of Bryophyta; Hutchinson & Co. Ltd.
- Campbell; 1939: The evolution of land plants; Stanford University.
- Spome, K.R. 1991. The Morphology of Pteridophytes.
- Parihar N.S. 1996 Biology and Morphology of Pteridophytes, Central Book Depot. Allahabad.
- Smith G. M.; Cryptogamic Botany Vol. II; Tata Mc Graw –Hill Publishing Co. Bombay -New Delhi
- Arnold C. A; An Introduction to Paleobotany; Tata Mc Graw –Hill Publishing Co. New Delhi.
- Stewart, W. N. and Raven G. W. 1993. Paleobotany and the Evolution of Plants. Cambridge University Press.
- Eames A J.; Morphology of Vascular Plants-Lower Groups. Tata Mc Graw -Hill Publishing Co. New Delhi
- Rashid A. 1999; An introduction to Pteridophytes; Vikas Publishing House Pvt. Ltd..
- Parihar; 1965: Pteridophyta; Central Book Depot. Allahabad.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

First Semester

Course MBT 103: Biology & Diversity of Gymnosperms

UNIT I: Introduction: Gymnosperms, the vesseless and fruitless seed plants; evolution of gymnsperms; complexity of female gametophytes.

UNIT II: Classification of gymnosperms and their distribution in India. Economic importance of gymnosperms

UNITIII: Gerenal account of pteridospermales, cycadeoidales and cordaitales. UNIT IV:Structure, reproduction and interrelationships of cycadales, ginkgoales and coniferales.

UNIT V: Structure, reproduction and interrelationships of ephedrales, welwitschiales and gnetales.

Suggested Readings

Bhatnagar, S.P. and Moitra, A; 1996: Gymnosperms. New Age International Pvt. Ltd., New Delhi.

Singh H.; 1978: Embryology of Gymnosperms, Encyclopedia of Plant Anatomy X. Gebruder Bortraeger, Berlin.

Spome K R; 1991: The Morphology of Gymnosperms; Hutchinson Univ. Library; London.

Foster A S. & Gifford E. M; Comparative morphology of vascular Plants; Vakils, Feffer, & Simons Private Ltd. Bombay. Chamberlain; Gymnosperms -Structure & Evolution; CBS Publishers & Distributors Delhi.

Shukla A C. & Mishra S. P.; Essentials of Paleobotany; Vikas Publishing House Pvt. Ltd. Delhi-Bombay-:6angalore-Calcutta-Kanpur .

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M. Sc. Botany (Semester System)

First Semester

Course MBT 104: Plant Ecology

- UNIT I: **Population Ecology:** Ecology & ecosystem: Definitions, Organization and components, Population & Environment; Population ecology, density & distribution, Natality, Mortality, Survivorship curves, Age structure & pyramids, Fecundity schedules, Life tables; Population growth – exponential and logistic curves; Intra specific competition and self regulation; r-and k-strategists.
- UNIT II: **Community organization:** Concepts of community and continuum; Analysis of community analytical and synthetic characters, Community coefficients and indices of diversity, interspecific association negative and positive associations; Concept of ecological niche; Concepts of biodiversity; evolution and differentiation of species – allopatric & sympatric speciation; ecads and ecotypes.
- UNIT III: **Ecosystem development and stability:** Temporal changes cyclic and non cyclic; Succession processes & types; Mechanism of succession facilitation, Tolerance and inhibition models; Concept of climax persistence resilience and resistance; Ecological perturbation natural and anthropogenic, Ecosystem restoration.
- UNIT IV: **Fate of energy in ecosystems:** Trophic organization and structure, Food chains & webs; energy flow pathways, Ecological efficiencies consumption, assimilation and production trophic; Primary production methods of measurement, Global patterns, Limiting factors.
- UNIT V: **Fate of matter in ecosystems:** Recycling pathways; Relationship between energy flow and recycling pathways; Nutrient exchange and cycling; Global biogeochemical cycles of C, N, P and S; Physical, chemical and Biological characteristics of soil.

Suggested Readings

Smith. R.L. 1996. Ecology and Field Biology. Harper Collins. New York. Muller-Dombois. D. and Ellenberg. H. 1974. Aims and Methods of Vegetation Ecology, Wiley, New York

Begon. M., Harper, J.L. and Townsend, C.R. 1996. Ecology. Blackwell Science. Cambridge.

Ludwig. J. and Reynolds. J.F. 1988. Statistical Ecology. John Wiley & Sons.

Odum. E.P. 1971. Fundamentals of Ecology. Saunders, Philadelphia.

Odum, E.P. 1983. Basic Ecology. Saunders, Philadelphia.

Barbour, M.G., Burk, J.H. and Pitts, W.O. 1987. Terrestrial Plant Ecology. Cummings Publication Company, California.

Kormondy, E.J. 1996. Concepts of Ecology. Prentice-Hall of India Pvt. Ltd., New Delhi.

Chapman, J.L. and Reiss, M.J. 1988. Ecology: Principles and Applications. Cambridge University Press, Cambridge, U.K.

Moldan, B. and Billharz, S. 1997. Sustainability Indicators. John Wiley & Sons, New York.

Treshow, M. 1985. Air Pollution and Plant Life. Wiley Interscience.

Heywood, V.H. and Watson, R.T. 1995. Global Biodiversity Assessment. Cambridge University Press.

Mason, C.F. 1991. Biology of Freshwater Pollution. Longman. '

Hill, M.K. 1997. Understanding Environmental Pollution. Cambridge University Press. .

Brady, N.C. 1990. The Nature and Properties of Soils. MacMillan.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

First Semester

Course MBT 105 : Practical-I

Practical List - I

Material A (Fungi)	-	06
Material B (Algae)	-	10
Bacterial staining	-	05
Comment upon plant disease	-	04
Spotting 01 to 05	-	10
Viva Voce	-	05
Sessional	-	10

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M.Sc. Botany (Semester System)

First Semester

Course MBT 106 : Practical-II

Practical List - II

Material A (Bryophyte)	-	05
Material B (Pteridophyte)	-	08
Material C (Gymnosperm)	-	10
Spotting 01 to 06	-	12
Viva Voce	-	05
Sessional	-	10

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Second Semester

Course MBT 201: Plant Development & Reproduction

- UNIT I: Unique features of plant development; differences between animal and plant development. Organization of shoot apical meristem (SAM); control of tissue differentiation, especially xylem and phloem; secretory ducts and laticifers. Wood development in relation to environmental factors.
- UNIT II: Leaf growth and differentiation. Organization of root apical meristem (RAM); cell fates and lineages; vascular tissue differentiation; lateral roots; root hairs. Root – microbe interaction.
- UNIT III: Vegetative options and sexual reproduction; flower development; genetics of floral organ differentiation; homeotic mutants in *Arabidopsis* and *Antirrhinum*; sex determination. Structure of anthers, microsporogenesis, role of tapetum, pollen development and gene expression.
- UNIT IV: Male sterility; pollen germination, pollen tube growth and guidance. Pollen storage, pollen allergy and pollen embryos. Ovule development, megasporogenesis; organization of embryo sac; structure of embryo sac cells.
- UNIT V: Floral characteristics; pollination mechanisms and vectors; breeding systems; structure of pistil; pollen stigma interactions; sporophytic and gametophytic self-incompatibility. Double fertilization. Endosperm development during early, maturation and desiccation stages; embryogenesis; storage proteins of endosperms and embryo. Polyembryony, apomixis. Dynamics of fruit growth; biochemistry and molecular biology of fruit maturation.

Suggested Readings

- Bhojwani, S.S. and Bhatnagar, S.P. 2000. The Embryology of Angiosperms (4th revised and enlarged edition). Vikas Publishing House, New Delhi.
- Burgess, J. 1985. An introduction to Plant Cell Development. Cambridge University Press, Cambridge.
- Fager, K. and Van der Pijl, L. 1979. The Principles of Pollination Ecology. Pergamon Press, Oxford.
- Fahn, A. 1982. Plant Anatomy. (3rd edition). Pergamon Press, Oxford.
- Fosket, D. E. 1994. Plant Growth and Development. A Molecular Approach. Academic Press, San Diego.
- Howell, S.H. 1998. Molecular Genetics of Plant Development, Cambridge University Press, Cambridge.
- Leins, P., Tucker, S.C. and Endress, P. K. 1988. Aspects of Floral Development. J. Cramer, Germany.
- Lyndon, R.F. 1990. Plant Development. The Cellular Basis. Unin Hyman. London.

Murphy, T. M. and Thompson, W. E. 1988 Molecular Plant Development. Prentice Hall, New Jersey.

Proctor, M. and Yeo, P. 1973. The Pollination of Flowers. William Collins Sons, London.

Raghavan, V. 1997. Molecular Embryology of Flowering Plants. Cambridge University Press, Cambridge.

Raghavan, V. 1999. Development Biology of Flowering Plants. Springer-verlag.

Atwell, B. J., Kriedemann, P. E. and Turnbull, C. G. N. (eds) 1999. Plants in Action: Adaptation in Nature, Performance in Cultivation, MacMillan Education, Sydney, Australia.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Second Semester

Course MBT 202: Morphology & Taxonomy of Angiosperms

- UNIT I: Morphology of stamens and carpels; carpel evolution. Morphology of inferior ovary; placentation types and their origin.
- UNIT II: The species concept: taxonomic hierarchy; species, genus, family and other categories; principles used in assessing relationships, delimitation of taxa and attribution of rank. Salient features of International code of Botanical Nomenclature.
- UNIT III: Taxonomic evidence: morphology, anatomy, palynology, embryology, cytology, phytochemistry, genome analysis and nucleic acid hybridization. Relevance of taxonomy to conservation.
- UNIT IV: Taxonomic tools: herbarium, floras, histological, cytological, phytochemical, serological, biochemical and molecular techniques; Computers and GIS. Local plant diversity and its socio-economic importance.
- UNIT V: Systems of angiosperm classification: phenetic versus phylogenetic systems; cladistics in taxonomy; relative merits and demerits of major systems of classification. Endemism, hot spots, hottest hot spots; plant explorations; invasions and introductions.

Suggested Readings

- Heywood & Moore, D.M; 1984: CWTent concept *in* Plant Taxonomy Academic Press.
- Banson, L.B.; 1957: Plant Classification, Heath & Co. Boston.
- Davis, P.R & Heywood, V.H 1973: Principles of Angiosperms and Taxonomy, Robert E. Kreiger Pub. Co. New York, USA
- Eames, A.I.; 1961: Morphology of Angiosperms, Mc-Graw Hill, New York.
- Jeffery, C.; 1968: An Introduction to Plant Taxonomy J. & H. Churchill Limited.
- Lawrence, G .H.M.; 1951: Taxonomy of Vascular Plants Macmillan, New York.
- Naik V. N.; 1984: Taxonomy of Angiosperms. Tata Mc-Graw Hill Pub. Co. Ltd. New Delhi.
- Porter, L.L.; 1959: Taxonomy of Flowering Plants. San Francisco. Radford~ A.E. Dickinson, W.C. Massey J.R and. Ben. C.R: 1974: VQ~llar Plant SYstematics, Harper & Row, New York

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Second Semester

Course MBT 203: Utilization & Conservation of Plant Resources

- UNIT I: **Plant Biodiversity:** Major Biomes of the world, Tropical rain & Seasonal Forests, Temperate rain & Seasonal forests, Boreal forests, Grasslands, Deserts; Aquatic Ecosystems, wetlands, Lakes & Ponds Streams & Rivers, Marine & Estuarine habitats.
- UNIT II: **Sustainable Development:** Resource utilization; Status & Utilization of Biodiversity; Sustainable development and utilization of resources from forest, Grassland and aquatic habitats; Food forage, Fodder, Timber & Non-wood forest products; Threats to quality & quantity of Resources due to overexploitation.
- UNIT III: **Strategies for conservation of resources:** Classifications of resources; Principles of conservation; *In-situ* conservation, sanctuaries, National parks, Biosphere reserves for wildlife conservation; Habitat conservation practices of conservation for forests, ranges, soil and water; Ex-situ conservation, botanical gardens, field gene banks, seed banks, in vitro repositories, cryo-banks.
- UNIT IV: **Pollution & Climate Change:** Air, Water and Soil pollution, Kinds, Sources, Quality parameters, Effects on structure & function of ecosystems; Management of pollution; Bioremediation; Climate changes sources, Trends & role of greenhouse gases, Effect of global warming on climate, Ecosystem processes & Biodiversity; Ozone layer & Ozone hole.
- UNIT V: **Resource monitoring:** Remote sensing concepts & Tools, Satellite remote sensing basics sensors, Visual & digital interpretation, EMR bands and their applications; Indian remote sensing program; Thematic mapping of resources; Application of remote sensing in Ecology & Forestry.

Suggested Readings

- Moldan, B. and Billharz, S. 1997. Sustainability Indicators. John Wiley & Sons, New York.
- Treshow, M. 1985. Air Pollution and Plant Life. Wiley Interscience.
- Heywood, V.H. and Watson, R.T. 1995. Global Biodiversity Assessment. Cambridge University Press.
- Mason, C.F. 1991. Biology of Freshwater Pollution. Longman. '
- Hill, M.K. 1997. Understanding Environmental Pollution. Cambridge University Press. .
- Brady, N.C. 1990. The Nature and Properties of Soils. MacMillan.
- Kothari, A 1997. Understanding Biodiversity: Life'Sustainability and Equity. Orient Longman.
- Kohli, R., Arya, K.S., Singh, P.H. and Dhillon, H.S. 1994. Tree Directory of Chandigarh. Lovedale Educational, New Delhi.
- Nair, M.N.B. et. al (Eds) 1998. Sustainable Management of Non-wood Forest Products. Faculty of Forestry, Universiti Putra Malaysia. 434004 PM Serdang, Selangor, Malaysia.

Paroda, R.S. and Arora, R.K. 1991. Plant Genetic Resources Conservation and Management. IPGRI (Publication) South Asia Office, C/o NBPGR, Pusa Campus, New Delhi.

Pimentel, D. and Hall, C.W. (eds) 1989. Food and Natural Resources. Academic Press, London-New York. .

Pinstrup-Andersen, P. et al. 1999. World Food Prospects: Critical Issues for the Early 21 st Century. International Food Policy Research Institute, Washington, D.C., USA.

Plant Wealth of India 1997. Special Issue of Proceedings Indian National Science Academy B-63.

Pulcknett, D.L., Smith, N.J.H., William, J.T. and Murti Annishetty, N. 1987. Gene Banks and World's Food. Princeton University Press, Princeton, New Jersey, USA.

Rodgers, N.A. and Panwar, H.S. 1988. Planning a Wildlife Protected Area Network in India. Vol. 1. The Report. Wildlife Institute of India, Dehradun.

Sahni, K.C. 2000. The Book of Indian Trees, 2nd edition. Oxford University Press, Mumbai.

Schery, R. W. 1972. Plants for Man. 2nd ed. Englewood Cliffs, New Jersey. Prentice Hall.

Sharma, O.P. 1996. Hill's Economic Botany (late Dr. A.F. Hills, adapted by O.P. Sharma) Tata McGraw Hill Co. Ltd., New Delhi. .. Swaminathan, M.S. and Kocchar, S.L. (Eds) 1989. Plants and Society. Macmillan, Publication Ltd., London.

Thakur, R.S., Puri, H.S. and Husain, A. 1989. Major Medicinal Plant of India. Central Institute of Medicinal and Aromatic Plants, CSIR, Lucknow.

Thomas, P. 2000. Tress: Their National History. Cambridge University Press, Cambridge.

Wagner, H., Hikino, H. and Farnsworth, N. 1989. Economic and Medicinal Plant Research. " Vols 1-3 Academic Press, London.

Walter, K.S. and Gillett, H.J. 1998. 1997 IUCN Red List of Threatened Plants. IUCN, the World Conservation Union. IUCN, Gland, Switzerland, and Cambridge, U.K.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Second Semester

Course MBT 204: Cell Biology of Plants

- UNIT I: Structural organization of the plant cell; specialized plant cell types. Structure and functions of cell wall; biogenesis; growth. Cytoskeleton: organization and role of microtubules and microfilaments; motor movements.
- UNIT II: Plasma membrane: structure, models and functions; sites for ATPases; ion carriers, channels and pumps; receptors. Structure of plasmodesmata, role in movement of molecules; comparison with gap junctions.
- UNIT III: Chloroplast: structure, genome organization, gene expression, nucleo-chloroplastic interactions; mitochondria: structure, genome organization, biogenesis. Plant vacuoles: tonoplast membrane, ATPases, transporters, as storage organelle. Other cell organelles: golgi apparatus, lysosomes, endoplasmic reticulum.
- UNIT IV: Nucleus: structure. Cell cycle: control mechanisms; role of cyclins and cyclin-dependent kinases; mechanisms of programmed cell death. Chromosome structure and packaging of DNA; euchromatin and heterochromatin; karyotype analysis and evolution; banding patterns; specialized types of chromosomes.
- UNIT V: Origin, meiosis and breeding behaviour of duplication, deficiency, inversion and translocation heterozygotes; origin, occurrence, production and meiosis of haploids, aneuploids and euploids; Origin and production of autopolyploids. Allopolyploids; types, genome constitution and analysis.

Suggested Readings

- Lewin, B. 2000, Genes VII Oxford University Press, New York.
- Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K., and Watson, J.D. Molecular Biology of the Cell. Garland Publishing: Inc., New York.
- Wolfe, S.L. 1993. Molecular and Cellular Biology, Wadsworth Publishing Co., California, USA
- Rost, T. et. al. 1998. Plant Biology, Wadsworth Publishing Co., California, U.S.A
- Krishanmurthy K V. 2000 Methods in Cell Wall Cytochemistry, CRC Press, Boca Raton, Florida U.S.A
- Buchanan, B.B. Groissem, W. and Jones, R.L. 2000. Biochemistry And Molecular Biology of Plants. American Society of Plant Physiologists, Maryland, USA
- De, D.N. 2000: Plant Cell Vacuoles: An Introduction. CSIRO Publication, Collingwood~ Australia.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Second Semester

Course MBT 205 : Practical-I

Practical List - I

- 1) To determine mitotic index in onion root dip cells.
- 2) Demonstration of special chromoscme.
- 3) Preparation of karyo type and ideogram.
- 4) Experiments on mutation.
- 5) Demonstration of aneuploidy, polyploidy, mendel's law, Neomendelism etc.
- 6) Study of root and snoot apices.
- 7) Study of different types of tissue.
- 8) Study of anomalous growth.
- 9) Anatomic, study of monocot and dicot root, stem and leaf.
- 10) Morphological study of modification of root, stem, and leaf.
- 11) Study phyllotaxis.
- 12) Study of different types of pollen grains and pollen germination.
- 13) Morphological study of different types of stomen and carpel.
- 14) Study of different types of placentation and ovules.
- 15) Study of different types of fruits.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Second Semester

Course MBT 206 : Practical-II

Practical List - II

- 1) To determine rate of transpiration by different method.
- 2) Comparison of rate of transpiration from two surface of leaves
- 3) Demonstration of osmosis, plasmolysis and deplasmolysis.
- 4) Determine effect of temperature/ pH/ toxin on permeability of cell membrane.
- 5) Determination of osmotic concentration of cell sap.
- 6) Study of physiological apparatus.
- 7) Preparation of normal, molal and molar solution.
- 8) Bioassay of plant hormones –auxins, ethylene, GA, ABA and cytokinin.
- 9) To determine stomatal index, stomatal frequency and percentage leave area open through stomata. Also, to study the effect of ABA on stomatal closure.
- 10) Vegetation analysis- frequency, density, dominance and abundance RF, RD, RA, RDOM by using different types of sampling unit-quadrates, line transect, belt transect, point center.
- 11) To determine diversity indices (Comparison of two community).
- 12) Determination IVI and preparation of phytograph.
- 13) Soil analysis.
- 14) Water analysis.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Third Semester

Course MBT 301: Plant Physiology

- UNIT I: Principles of thermodynamics, free energy and chemical potential, redox reactions, structure and functions of ATP, Plant water relations, mechanisms of water transport through xylem, root-microbe interactions in facilitating nutrient uptake. Membrane transport proteins.
- UNIT II: Phloem transport; phloem loading and unloading, passive and active solute transport. Signal transduction; overview, receptors and proteins, phospholipids signaling, role of cyclic nucleotides, calcium-calmodulin cascade. Specific signaling mechanisms, for example, two-component sensor regulator system in bacteria and plants. Sucrose-sensing mechanism.
- UNIT III: Plant growth regulators and elicitors: Physiological effects and mechanism of action of auxins, gibberellins, cytokinins, ethylene, abscisic acid, brassinosteroids, polyamines, jasmonic acid, and salicylic acid. Hormone receptors, signal transduction and gene expression.
- UNIT IV: Flowering process: photoperiodism and its significance, endogenous clock and its regulation. Floral induction and development. Phytochromes and cryptochromes, their photochemical and biochemical properties. Molecular mechanism of action of photomorphogenetic receptors, signaling and gene expression. Role of vernalization.
- UNIT V: Stress physiology: Plant responses to biotic and abiotic stress. Water deficit and drought resistance. Salinity stress and resistance. Concepts of freezing, heat and oxidative stresses.

Suggested readings

- Hopkins, W.G. 1995. Introduction to Plant Physiology. John Wiley & Sons, Inc., New York, USA.
- Lodish, H., Berk, A., Zipursky, S.L., Matsudaira, P., Baltimore, D. and Darnell, J. 2000. Molecular cell Biology (fourth edition). W.H. Freeman and Company, New York USA.
- Moore, T.C. 1989. Biochemistry and Physiology of Plant Hormones (second edition). Springer-Verlag, New York. USA.
- Nobel, P.S. 1999. Physiochemical and Environmental Plant Physiology (second edition). Academic Press, San Diego, USA.
- Salisbury FB and Ross CW 1991 Plant Physiology IV edition Wdsworth Publishing co. California usa.
- Taiz I and Zeiger E 1998 Plant Physiology II Edition. Sinauer Associates Inc. Publisher MS.
- Dennis DT and Terpin DH Lefevre DD and Layzell DV 1997 Plant Metabolism II Ed. Longman England.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Third Semester

Course MBT 302: Plant Biochemistry & Metabolism

- UNIT I: Fundamentals of enzymology: general aspects, allosteric mechanism, regulatory and active sites, isozymes, kinetics of enzymatic catalysis, Michaelis-Menten equation and its significance, Mechanism of enzyme action.
- UNIT II: Photochemistry and photosynthesis: General concepts and historical background, evolution of photosynthetic apparatus, photosynthetic pigments and light-harvesting complexes. Photooxidation of water, mechanism of electron and proton transport, Carbon assimilation; Calvin cycle, photorespiration and its significance, C₄-cycle, CAM pathway, physiological and ecological considerations. Biosynthesis of starch and sucrose.
- UNIT III: Respiration and lipid metabolism: Overview of plant respiration, glycolysis, TCA cycle, electron transport and ATP synthesis. Oxidative pentose phosphate pathway, glyoxylate cycle, alternative oxidase system.
- UNIT IV: Structure and functions of lipids, fatty acid biosynthesis, synthesis of membrane lipids, structural lipids and storage lipids and their catabolism. Sulphate uptake, transport and assimilation.
- UNIT V: Nitrogen fixation, nitrogen and sulphur metabolism: Overview, biological nitrogen fixation, nodule formation and nod factors. Mechanism of uptake and reduction, ammonium assimilation.

Suggested readings

- Hopkins, W.G. 1995. Introduction to Plant Physiology. John Wiley & Sons, Inc., New York, USA.
- Lodish, H., Berk, A., Zipursky, S.L., Matsudaira, P., Baltimore, D. and Darnell, J. 2000. Molecular cell Biology (fourth edition). W.H. Freeman and Company, New York USA.
- Moore, T.C. 1989. Biochemistry and Physiology of Plant Hormones (second edition). Springer-Verlag, New York. USA.
- Nobel, P.S. 1999. Physiochemical and Environmental Plant Physiology (second edition). Academic Press, San Diego, USA.
- Salisbury FB and Ross CW 1991 Plant Physiology IV edition Wdsworth Publishing co. California usa.
- Taiz I and Zeiger E 1998 Plant Physiology II Edition. Sinauer Associates Inc. Publisher MS.
- Dennis DT and Terpin DH Lefevre DD and Layzell DV 1997 Plant Metabolism II Ed. Longman England.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Third Semester

Course MBT 303: Genetics & Cytogenetics

- UNIT I: Genetics of prokaryotes and eukaryotic organelles; genetic recombination in phage; mapping the bacteriophage genome; genetic transformation, conjugation and transduction in bacteria. Genetics of mitochondria and chloroplasts; cytoplasmic male sterility.
- UNIT II: Genetic recombination and genetic mapping; Recombination, independent assortment and crossing-over, molecular mechanism of recombination. Chromosome mapping, linkage groups, genetic markers, construction of molecular maps, somatic cell genetics- an alternative approach to gene mapping.
- UNIT III: Mutations: spontaneous and induced mutations, physical and chemical mutagens, molecular basis of gene mutations. Transposable elements in prokaryotes and eukaryotes. Mutations induced by transposons, site-directed mutagenesis. DNA damage and repair mechanisms.
- UNIT IV: Cytogenetics of aneuploids and structural heterozygotes, effect of aneuploidy on phenotype in plants, transmission of monosomics and trisomics and their use in chromosome mapping of diploid and polyploidy species. Complex translocation heterozygotes. Robertsonian translocations. B-A translocations.
- UNIT V: Molecular Cytogenetics; Nuclear DNA content, c-value paradox, *cot* curve and its significance, restriction mapping – concept and techniques, multigene families and their evolution. Transfer of whole genome, examples from wheat and *Brassica*; genetic basis of inbreeding and heterosis; exploitation of hybrid vigour.

Suggested Readings

- Alberts, B. Bray, D. Lewis, J. Raff: M. Roberts, K. and Watson, J. D. 1989 Molecular Biology of the Cell (2nd edition). Garland Publishing Inc., New York. U. S. A.
- Atherly, AG. Girton, J.R and Mc Donald, J.E. 1999. The Science of Genetics: Saunders College Publishing, Fort Worth, U.S.A.
- Burnham, C.R 1962. Discussions in Cytogenetics, Burgess Publishing Co. Minnesota.
- Busch. H. and Rothblum. L. 1982. Volume X. The Cell Nucleus rDNA Part A. Academic Press.
- Hartl, D.L. and Jones, E. W. 1998. Genetics: Principles and Analysis (4th edition). Jones & Bartlett Publishers, Massachusetts, USA
- Khush, G.S. 1973. Cytogenetics of Aneuploids. Academic Press, New York, London.
- Karp, G. 1999 Cells and Molecular Biology: Concepts and Experiments John Wiley & Sons, Inc. U. S.A
- Lewin: B. 2000 Gene VII. Oxford University Press, New York, U.S.A
- Lewis, R. 1997. Human Genetics: Concepts and Applications. (2nd edition). WCB McGraw Hill, U.S.A

Malacinski, G. M. and Freifelder, D. 1998. Essentials of Molecular Biology (3^M edition). Jones and Barlett Publishers, Inc. London.

Russel, P.J. 1998. Genetics (5th edition). The Benjamin / Cummings Publishing Company Inc., U. S. A

Snustead, D. P. and Simmons, M. J. 2000. Principles of Genetics (2nd edition). Jhon Wiley & Sons Inc., U.S.A

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Third Semester

Course MBT 304: Molecular Biology of Plants

- Unit I: DNA structure; A, B and Z forms; replication; damage and repair; transcription; plant promoters and transcription factors; splicing; messenger RNA transport; ribosomal RNA biosynthesis.
- Unit II: Gene structure and expression; genetic fine structure; cis-trans test; fine structure analysis of eukaryotes; introns and their significance; RNA splicing; regulation of gene expression in prokaryotes and eukaryotes.
- Unit III: Ribosomes: structure and site of protein synthesis; mechanism of translation, initiation, elongation and termination; structure and role of transfer RNA; protein sorting; targeting of proteins to organelles.
- Unit IV: Cell cycle and apoptosis, control mechanisms; role of cyclins and cyclin dependent kinases; retinoblastoma and E2F proteins; cytokinesis and cell plate formation; mechanism of programmed cell death.
- Unit V: Immunotechniques, In situ hybridization – concepts and techniques, physical mapping of genes on chromosomes. In situ hybridization to locate transcript in cell types; FISH; Flow cytometry and confocal microscopy.

Suggested Readings

- Lewin, B. 2000, Genes VII, Oxford University Press, New York.
- Alberts, B., Bray, D., Lewis, J., Raff, M., Roberts, K., and Watson, J.D. 1999. Molecular Biology of the Cell. Garland Publishing: Inc., New York.
- Wolfe, S.L. 1993. Molecular and Cellular Biology, Wadsworth Publishing Co., California, USA.
- Rost, T. et al. 1998. Plant Biology, Wadsworth Publishing Co., California, U.S.A
- Krishanmurthy K. V. 2000 Methods in Cell Wall Cytochemistry, CRC Press, Boca Raton, Florida U.S.A
- Buchanan, B.B. Gruissem, W. and Jones, R.L. 2000. Biochemistry And Molecular Biology of Plants. American Society of Plant Physiologists, Maryland, USA
- De, D.N. 2000 Plant Cell Vacuoles: An Introduction. CSIRO Publication, Collingwood, Australia.
- Kleinsmith, L.J. and Kish, V.M. 1995. Principles of Cell and Molecular Biology (2nd Edition). Harper Collins College Publishers, New York, USA
- Lodish H. Bert, A., Zipursky, S.L., Matsudaira, P., Baltimore, D. and Darnell, J. 2000. Molecular Cell Biology, W.H. Freeman and Co., New York, U.S.A
- Pollard, T. S. and Eashaw, W. C. 2002 Cell Biology. Saunders, Philadelphia, U.S.A
- Twyman, R. M. 2003. Advanced Molecular Biology. Viva Books Private Ltd. New Delhi.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Third Semester

Course MBT 305 : Practical-I

Practical List - I

- 1) Taxonomic description of various plant species (from local flora) based on syllabus.
- 2) Techniques of herbarium preparation, Preparation of artificial key.
- 3) Isolation of DNA and RNA from plant material.
- 4) Demonstration of DNA sequencing by sangers dideoxy method.
- 5) Isolation of DNA and preparation by a spectrophotometric method.
- 6) Isolation of Genomic DNA from plant or animal tissue using CTAB method.
- 7) Isolation of chloroplast and SDS –PAGE. Profile of demarcate the two subunits of Rubisco.
- 8) Demonstration of gel electrophoresis.
- 9) Demonstration of molecular markers.
- 10) Plant breeding techniques (Emasculation).
- 11) Isolation of DNA and Preparation of Cot curve.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Third Semester

Course MBT 306 : Practical-II

Practical List - II

- 1) Determine rate of respiration in germinating/young buds by Ganong's respirometer.
- 2) Determine effect of light intensity/ wavelength on rate of photosynthesis.
- 3) Extraction of chloroplast pigments from leaves and preparation of the absorption spectrum of chlorophylls and carotenoids.
- 4) To determine the chlorophyll a/ chlorophyll b ratio in C3 and C4 plants.
- 5) Effect of time and enzyme concentration on the rate of reaction of enzyme C e.g. acid phosphatases, nitrate reductase.
- 6) Effect of substrate concentration on activity of any enzyme C (catalase, amylase).
- 7) Demonstration of the substrate inducibility of the enzyme nitrate reductase.
- 8) Determination of succinate dehydrogenase activity/ its kinetic and sensitivity to inhibitors.
- 9) Separation of isoenzymes of esterase, peroxidases by native polyacrylamide gel electrophoresis.
- 10) Biochemical test.
- 11) Forest grass land and aquatic resources (Plants) food, fibers, fodder, vegetable, timber.
- 12) Study of the following through maps:
 - a) Status of plant biodiversity in India.
 - b) Center of origin
 - c) Sanctuaries and national park
 - d) Hot spots.
- 13) Comparison analysis of soil and water from polluted and non polluted sites.
- 14) Determination leaf injury index of samples collected from polluted and non polluted sites.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester

Course MBT 401: Plant Cell, Tissue & Organ Culture

UNIT I: Plant cell and tissue culture: general introduction, history, scope, concept of cellular differentiation and totipotency.

UNIT II: Techniques of tissue culture. Organ culture – meristem, anther and embryo. In vitro fertilization.

UNIT III: Organogenesis and adventive embryogenesis; fundamental aspects of morphogenesis, somatic embryogenesis and androgenesis. Mechanisms, techniques and utility.

UNIT IV: Somatic hybridization, protoplast isolation, fusion and culture, hybrid selection and regeneration; possibilities and achievements and limitations of protoplast research.

UNIT V: Application of plant tissue culture; clonal propagation; artificial seeds; production of hybrids, somaclones and somaclonal variation; production of secondary metabolites/natural products; cryopreservation and germplasm storage.

Suggested Reading

Butenko, R.G. 2000. Plant Cell Culture. University Press of Pacific.

Collin, H.A. and Edwards, S. 1998. Plant Cell Culture. Bios Scientific Publishers, Oxford, UK.

Dixon, R.A. (Ed.) 1987. Plant Cell Culture: A Practical Approach. IRL Press, Oxford.

Gelvin, S.B. and Schilperoort, R.A. (Eds), 1994. Plant Molecular Biology Manual, 2nd edition, Kluwer Academic Publishers, Dordrecht, The Netherlands.

George, E.F. 1993. Plant Propagation by Tissue Culture. Part 1. The Technology, 2nd edition. Exegetics Ltd., Edington, UK.

George, E.F. 1993. Plant Propagation by Tissue Culture. Part 2. In Practice, 2nd edition. Exegetics Ltd., Edington UK.

Glick, B.R. and Thompson, J.E. 1993. Methods in Plant Molecular Biology and Biotechnology. GRC Press, Boca Raton, Florida.

Glover, D.M. and Hames, B.D. (Eds), 1995. DNA Cloning 1 : A Practical Approach; Core Techniques, 2nd edition. PAS, IRL Press at Oxford University Press, Oxford.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester

Course MBT 402: Biotechnology & Genetic Engineering

- UNIT I: Biotechnology and Genetic Engineering: basic concepts, principles and scope. Intellectual Property Rights – possible ecological risks and ethical concerns.
- UNIT II: recombinant DNA technology; gene cloning – principles and techniques; construction of genomic/ cDNA libraries; choice of vectors; DNA synthesis and sequencing, polymerase chain reaction. DNA fingerprinting
- UNIT III: Genetic engineering of plants, aims, strategies for development of transgenics (with suitable examples); *Agrobacterium* – the natural genetic engineer; T-DNA and transposon mediated gene tagging; chloroplast transformation and its utility.
- UNIT IV: Microbial genetic manipulation; bacterial transformation; selection of recombinants and transformants; genetic improvements of industrial microbes and nitrogen fixers; fermentation technology.
- UNIT V: Genomics and Proteomics; genetic and physical mapping of genes; molecular markers for introgression of useful traits; artificial chromosomes; high throughput sequencing; genome projects; bioinformatics; functional genomics; microarrays; protein profiling and its significance.

Suggested Reading

- Gelvin, S.B. and Schilperoort, R.A. (Eds), 1994. Plant Molecular Biology Manual, 2nd edition, Kluwer Academic Publishers, Dordrecht, The Netherlands.
- Glick, B.R. and Thompson, J.E. 1993. Methods in Plant Molecular Biology and Biotechnology. GRC Press, Boca Raton, Florida.
- Glover, D.M. and Hames, B.D. (Eds), 1995. DNA Cloning 1 : A Practical Approach; Core Techniques, 2nd edition. PAS, IRL Press at Oxford University Press, Oxford.
- Brown, T.A. 1999. Genomes. John Wiley & sons, Singapore
- Callow, J.A., Ford-Lloyd, B.V. and Newbury, H.J. 1997. Biotechnology and Plant Genetic Resources: Conservation and Use. CAB International, Oxon, UK.
- Jolles, O. and Jornvall, H. (eds.) 2000. Proteomics in Functional Genomics. Birkhauser Verlag, Basel, Switzerland.
- Primrose, S.B. 1995. Principles of Genome Analysis. Blackwell Science Ltd., Oxford, UK

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester Course MBT 403: Limnology

Unit-1	General characteristics of lakes and classification of lakes. Definition depth of lakes. Retention and replacement of water in lakes, origin of lakes. Zonation of lakes.
Unit-2	Lake Flora Phytoplankton-classification of phytoplankton, phytoplankton flora, special distribution of phytoplankton, seasonal distribution and species composition of phytoplankton. Algal blooms effects of salinity and climatic stresses on the distribution of phytoplankton. Phytobenthos-classification
Unit-3	Lake Flora-Higher Plants. Categories of aquatic higher plants, zonation of rooted higher plants, some peculiarities of aquatic higher plants. Lake Bacteria-occurrence, characteristics and importance.
Unit-4	Physical factors affecting the flora. Light, temperature, waves and currents chemical factors affecting the flora-oxygen, carbon, dioxide, nutrients hydrogen ion, concentration.
Unit-5	Biotic relationship and interaction among organisms. Symbiosis, competition among algae, Parasitism of algae, predation of algae, impact of human being on algae.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester

Course MBT 404[I]: Industrial Microbiology (Elective)

Unit-1	Basic techniques in microbiology - Microscopy, staining techniques, Culture, Nutrition and growth of microorganisms. Replication and structure of viruses & other a cellular microorganisms, prokaryotic microorganisms, classification and diversity of Bacteria, Eukaryotic microorganisms.
Unit-2	Food Microbiology: Food spoilage, Food preservation methods, Microbiological production of food such as fermented products, alcoholic beverages, vinegar. Fermented vegetables. Single cell protein production in industry, fermented dairy products and uses.
Unit-3	Fermentation Industry: Selection of micro-organisms, Techniques and quality control, Production of antibiotics, steroids, Human proteins, Vaccines and vitamins. Survey of microorganisms of industrial uses. Production of organic acids, amino acids, Enzymes, Solvents and fuels.
Unit-4	Recovery of minerals by using microbes, Oil recovery, Bio deterioration, Mushroom culture, Biotech products including human insulin, Microbial Growth-Environmental influences, Physical control, Chemical control & Antibiotic controls.
Unit-5	Water quality in industry: Bacteriological safety of potable water, water quality analysis, importance of BOD. Biodegradation of wastes and pollutants, Primary, Secondary and Tertiary Sewage treatments.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester

Course MBT 404[II]: Taxonomy of Angiosperms (Elective)

UNIT -1

Nomenclature I.C.B.N. Principles, Rules, Recommendation, Articles, Typification, Principle of priority, Effective and Valid publication, citation of authority synonyms and homonyms.

The species concept Taxonomic hierarchy, topological biological delimitation of Taxa and attribution of rank, salient features of the International code of botanical nomenclature.

Plant speciation- Allopatric, sympatric, hybrid Apomictic Abrupt and phylletic, speciation, Mechanism of Reproductive isolation Taxonomic tools-herbarium Floras, Botanical garden, Botanical survey of India. Modern trends in Taxonomy-Morphology, Anatomy Palynology, Embryology, Cytology, Ecology, Phytochemistry, Genomic analysis Nucleic acid hybridization.

UNIT-II

Fossils Angiosperms , origin and evolution of angiosperms Endangered plants of m,p System of angiosperms classification phenetic versus phylogenetic system, cladistic in taxonomy. Principle, outline, merits and demerits of Bentham and hooker, Englar & Prantle, Hutchinson, Conquest, APG system. Morphology and evolutionary trends in angiosperms .Morphology of carpel, & stamen Monographic nature of flower. Calculation of similarity coefficients & preparation of dendrogram.

UNIT-III

Systematic account of following families [Salient feature ,Morphological diversity, Phylogenetic relationship and Economic importance] Ranunculaceae, Magnoliaceae, Annonaceae, Nymphaeaceae, Papaveraceae, Fumariaceae, Cruciferae (brassicaceae), Podostemaceae, Rosaceae, Linaceae, Capparidaceae, Violaceae, Caryophyllaceae, Dipterocarpaceae, Malvaceae, Tiliaceae , Sterculiaceae, Rutaceae, Meliaceae, Rhamnaceae, Sapindaceae, Anacardiaceae, Salvadoraceae, Leguminosae, Lythraceae, Myrtaceae, Onagraceae, Cucurbitaceae, Apiaceae, Tamaricaceae, Caricaceae Cactaceae,

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UNIT IV

Systematic account of Following Families[Salient feature, Morphological diversity , Phylogenetic relationship and Economic importance]Rubiaceae, Asteraceae, Primulaceae,Sapotaceae,Ileaceae Apocynaceae, Asclepiadaceae, Boraginaceae, Convolvulaceae, Solanaceae ,Pedaliaceae, Lentibulariaceae, Scrophulariaceae, Bignoniaceae, Acanthaceae,Verbinaceae, Labiatae, Amaranthaceae, Chenopodiaceae, Polygonaceae, Moraceae, Casuarinaceae, Combritaceae, Nyctaginaceae Bombacaceae, Urticaceae, Oxalidaceae, Moringaceae, Oliaceae, Malpighiaceae, Gentianaceae,Vitaceae

UNIT-V

Monocotyledons families and Phylogeny - lismaceae,Hydrochariaceae, Poaceae, Cyperaceae, Comelinaceae Arecaceae, Liliaceae Amaryllidaceae, Araceae Iridaceae, Musaceae, Zingiberaceae, Cannaceae, Orchidaceous Typhaceae, General characteristic phylogeny of following orderRanales, Amentiferae,Tubiflorae,Santalles, Centrospermatales, Halobiales, Floristic Works with reference to m.p.

BOOKS:-

- 1) Verma B.K.(2011)-Introduction to Taxonomy of Angiosperms.
- 2) Eames, A.J.(1961) Morphology of Angiosperms Mc-Graw Hill, New York.
- 3) Rendle. A.B. (1997) The Classification of flowering plants, Vol-1&II
- 4) Takhtajan, A.L.(1969) Flowering Plants- Origin and dispersal.
- 5) Lawrence,G.H.M. (1951)Taxonomy of Vascular plants MMacmillian, New York.

BHABHA UNIVERSITY, BHOPAL

M. Sc. Botany (Semester System)

Fourth Semester

Course MBT 404[III]: Bioinformatics and Computer Application & Biostatistics (Elective)

Unit-1	Bioinformatics - Objectives and scope of Bioinformatics, Data-base Search creation, Preparation and management, Use of currently important Databases in bioinformatics. Modern strategies of literature search, Significance of keywords in Literature search, use of e-mail and internet. And retrieving information on biotechnology.
Unit-2	Introduction to computer fundamentals, permanent storage of number systems, decimal to binary and vice-versa; binary coded Decimal number; MS - DOS, MS - WORD, MS - EXCEL. Application of computer in biostatistical problems. RDBMS, Definition of relational database, basic SQL prompts; retrieval sorting indexing and merging; Programming logic & techniques, Algorithm, flowcharts.
Unit-3	Computer in biology : Sequence data bases; sequence analysis of proteins and nucleic acids, structure prediction, simple molecular Modeling, sample graph plotting. Networking of computer, need and advantage, overview of Indian network Such as NICNET, INFLIBNET, Detailed study of Internet.
Unit-4	Computer oriented numerical methods, solution of simple algebraic and Transcendental equations. Solution of linear simultaneous equation, Numerical integration. Partial fraction, theory of quadratic equations, permutation and Combination, Binomial theorem, Finite and infinite series, Determinant. Differential and integral calculus - Theory and rules, methods of Differentiation and integration, Definite integral, Simple differential Equation.
Unit-5	Basic statistical concepts: Population and samples, classification, Summarization and presentation of data, Frequency distribution, Measures Of location and dispersion, Statistical concepts: Correlation and regression, probability models, Nomial and bi-nomial distributions, normalizing transformation and test Of inference, Analysis of variance.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Fourth Semester

Course MBT 405 : Practical-I

Practical List - I

- 1) Quantitation of nuclear acid or chlorophyll by a spectro-photometric method.
- 2) Demonstration of azrose gel electrophoresis and visualization by Et. Br. Staining.
- 3) Isolation of Chloroplast and SDS-PAGE. Profile of proteins two demarcate the two subunits of Rubisco.
- 4) Density gradient separation of cell organelles.
- 5) Demonstration of SEM and TEM.
- 6) Data presentation through computer.
- 7) Problems based on statistics.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Botany (Semester System)

Fourth Semester

Course MBT 406 : Practical-II

Practical List - II

- 1) Study of maceration of plant tissue.
- 2) Study of nonwood, Forest produce.
- 3) Study of phytogeographical region of India.
- 4) Study of grassland of M.P and India.
- 5) Study of forest types of M.P and India.
- 6) Study of Biosphere reserve.
- 7) Identification of medicinal plants on the basis of stomatal frequency.
- 8) Extraction of dyes from Ethnobotanical important plant.
- 9) Morphological study of Ethnobotanical plants.